My Senior Project

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```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   4.0.0     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
understat_per_game <- read_csv("/Users/abdullah/Downloads/archive (5)/understat_per_game.csv")
```

```
## Rows: 24580 Columns: 29
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr   (4): league, h_a, result, team
## dbl  (24): year, xG, xGA, npxG, npxGA, deep, deep_allowed, scored, missed, x...
## dttm  (1): date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
glimpse(understat_per_game)
```

```
## Rows: 24,580
## Columns: 29
## $ league       <chr> "Bundesliga", "Bundesliga", "Bundesliga", "Bundesliga", "~
## $ year         <dbl> 2014, 2014, 2014, 2014, 2014, 2014, 2014, 2014, 2014, 201~
## $ h_a          <chr> "h", "a", "h", "a", "h", "a", "h", "h", "a", "h", "a", "h~
## $ xG           <dbl> 2.5701200, 1.5032800, 1.2298700, 1.0351900, 3.4828600, 3.~
## $ xGA          <dbl> 1.1984200, 1.3079500, 0.3101660, 0.2031180, 0.4028440, 0.~
## $ npxG         <dbl> 2.5701200, 1.5032800, 1.2298700, 1.0351900, 3.4828600, 3.~
## $ npxGA        <dbl> 1.1984200, 1.3079500, 0.3101660, 0.2031180, 0.4028440, 0.~
## $ deep         <dbl> 5, 10, 13, 6, 23, 27, 14, 16, 5, 11, 13, 12, 19, 3, 13, 3~
## $ deep_allowed <dbl> 4, 1, 3, 2, 2, 0, 5, 0, 7, 3, 1, 6, 1, 1, 0, 1, 1, 3, 1, ~
## $ scored       <dbl> 2, 1, 2, 0, 4, 2, 4, 6, 0, 2, 4, 4, 1, 1, 4, 2, 2, 1, 1, ~
## $ missed       <dbl> 1, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 4, 1, ~
```

```
## $ xpts         <dbl> 2.3486, 1.5143, 2.1588, 2.1367, 2.9287, 2.8138, 2.7325, 2~
## $ result       <chr> "w", "d", "w", "d", "w", "w", "w", "w", "d", "w", "w", "w~
## $ date         <dttm> 2014-08-22 19:30:00, 2014-08-30 17:30:00, 2014-09-13 14:~
## $ wins         <dbl> 1, 0, 1, 0, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 1, 0, 0, ~
## $ draws        <dbl> 0, 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, ~
## $ loses        <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, ~
## $ pts          <dbl> 3, 1, 3, 1, 3, 3, 3, 3, 1, 3, 3, 3, 3, 3, 3, 3, 3, 0, 1, ~
## $ npxGD        <dbl> 1.3717000, 0.1953300, 0.9197040, 0.8320720, 3.0800160, 2.~
## $ ppda_coef    <dbl> 9.625000, 4.756098, 5.060606, 4.423077, 4.250000, 5.44000~
## $ ppda_att     <dbl> 231, 195, 167, 115, 170, 136, 187, 109, 126, 192, 153, 14~
## $ ppda_def     <dbl> 24, 41, 33, 26, 40, 25, 17, 30, 20, 31, 25, 25, 26, 22, 2~
## $ oppda_coef   <dbl> 21.850000, 17.695652, 16.961538, 9.446809, 44.800000, 44.~
## $ oppda_att    <dbl> 437, 407, 441, 444, 448, 485, 419, 432, 376, 402, 408, 34~
## $ oppda_def    <dbl> 20, 23, 26, 47, 10, 11, 8, 15, 14, 30, 49, 38, 25, 32, 32~
## $ team         <chr> "Bayern Munich", "Bayern Munich", "Bayern Munich", "Bayer~
## $ xG_diff      <dbl> 0.570120, 0.503280, -0.770130, 1.035190, -0.517140, 1.469~
## $ xGA_diff     <dbl> 0.1984200, 0.3079500, 0.3101660, 0.2031180, 0.4028440, 0.~
## $ xpts_diff    <dbl> -0.6514, 0.5143, -0.8412, 1.1367, -0.0713, -0.1862, -0.26~
```

Simple Linear Regression

```r
model1 <- lm(scored ~ npxGD, data = understat_per_game)
summary(model1)
```

```
##
## Call:
## lm(formula = scored ~ npxGD, data = understat_per_game)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.8289 -0.7912 -0.1324  0.6427  6.6282
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 1.341009   0.006883   194.8   <2e-16 ***
## npxGD       0.513745   0.005708    90.0   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.079 on 24578 degrees of freedom
## Multiple R-squared:  0.2479, Adjusted R-squared:  0.2478
## F-statistic:  8100 on 1 and 24578 DF,  p-value: < 2.2e-16
```

Multiple Linear Regression

```r
model2 <- lm(scored ~ xG_diff + xGA_diff + npxGD, data = understat_per_game)
summary(model2)
```

```
##
```

```
## Call:
## lm(formula = scored ~ xG_diff + xGA_diff + npxGD, data = understat_per_game)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.2536 -0.4163 -0.0741  0.3346  3.1875
##
## Coefficients:
##              Estimate Std. Error  t value Pr(>|t|)
## (Intercept)  1.311339   0.003605  363.726  < 2e-16 ***
## xG_diff     -0.966725   0.003786 -255.312  < 2e-16 ***
## xGA_diff     0.019238   0.003786    5.081 3.79e-07 ***
## npxGD        0.516142   0.002987  172.789  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5646 on 24576 degrees of freedom
## Multiple R-squared:  0.7941, Adjusted R-squared:  0.794
## F-statistic: 3.159e+04 on 3 and 24576 DF,  p-value: < 2.2e-16
```

Polynomial Regression

```r
model3 <- lm(scored ~ xG_diff + I(xG_diff^2), data = understat_per_game)
summary(model3)
```

```
##
## Call:
## lm(formula = scored ~ xG_diff + I(xG_diff^2), data = understat_per_game)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.4541 -0.6013 -0.1579  0.4388  5.3348
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.165322   0.005927  196.62   <2e-16 ***
## xG_diff      -0.870889   0.005711 -152.49   <2e-16 ***
## I(xG_diff^2)  0.163808   0.003358   48.78   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8025 on 24577 degrees of freedom
## Multiple R-squared:  0.584,  Adjusted R-squared:  0.584
## F-statistic: 1.725e+04 on 2 and 24577 DF,  p-value: < 2.2e-16
```

Linear Regression with Pressing Variables

```r
model4 <- lm(scored ~ xG_diff + ppda_coef + oppda_coef, data = understat_per_game)
summary(model4)
```

```
##
## Call:
## lm(formula = scored ~ xG_diff + ppda_coef + oppda_coef, data = understat_per_game)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.0814 -0.5891 -0.1334  0.4396  5.1249
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.1480120  0.0148043   77.55   <2e-16 ***
## xG_diff     -0.9766003  0.0053544 -182.39   <2e-16 ***
## ppda_coef   -0.0202945  0.0008375  -24.23   <2e-16 ***
## oppda_coef   0.0351889  0.0008358   42.10   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7969 on 24576 degrees of freedom
## Multiple R-squared:  0.5898, Adjusted R-squared:  0.5898
## F-statistic: 1.178e+04 on 3 and 24576 DF,  p-value: < 2.2e-16
```

Full Multiple Regression

```r
model5 <- lm(scored ~ xG_diff + xGA_diff + npxGD + ppda_coef + oppda_coef +
               ppda_att + ppda_def + oppda_att + oppda_def,
             data = understat_per_game)
summary(model5)
```

```
##
## Call:
## lm(formula = scored ~ xG_diff + xGA_diff + npxGD + ppda_coef +
##     oppda_coef + ppda_att + ppda_def + oppda_att + oppda_def,
##     data = understat_per_game)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.3849 -0.4141 -0.0715  0.3317  3.1780
##
## Coefficients:
##               Estimate Std. Error  t value Pr(>|t|)
## (Intercept)  1.0373710  0.0324773   31.941  < 2e-16 ***
## xG_diff     -0.9634701  0.0037809 -254.828  < 2e-16 ***
## xGA_diff     0.0243800  0.0037809    6.448 1.15e-10 ***
## npxGD        0.5112875  0.0032692  156.394  < 2e-16 ***
## ppda_coef    0.0057364  0.0012933    4.435 9.23e-06 ***
## oppda_coef   0.0054596  0.0012933    4.221 2.44e-05 ***
## ppda_att     0.0002153  0.0000783    2.750  0.00597 **
## ppda_def     0.0012190  0.0008222    1.483  0.13817
## oppda_att    0.0003895  0.0000783    4.974 6.60e-07 ***
## oppda_def   -0.0009222  0.0008222   -1.122  0.26201
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 0.5608 on 24570 degrees of freedom
## Multiple R-squared:  0.7969, Adjusted R-squared:  0.7969
## F-statistic: 1.071e+04 on 9 and 24570 DF,  p-value: < 2.2e-16
```

Log-Transformed Linear Model

```r
model6 <- lm(log(scored + 1) ~ xG_diff + xGA_diff + npxGD,
             data = understat_per_game)
summary(model6)
```

```
##
## Call:
## lm(formula = log(scored + 1) ~ xG_diff + xGA_diff + npxGD, data = understat_per_game)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.0794 -0.1844  0.0009  0.1799  1.1377
##
## Coefficients:
##              Estimate Std. Error  t value Pr(>|t|)
## (Intercept)  0.701198   0.001755  399.438  < 2e-16 ***
## xG_diff     -0.396850   0.001844 -215.250  < 2e-16 ***
## xGA_diff     0.006136   0.001844    3.328 0.000875 ***
## npxGD        0.207647   0.001454  142.765  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2749 on 24576 degrees of freedom
## Multiple R-squared:  0.7302, Adjusted R-squared:  0.7302
## F-statistic: 2.217e+04 on 3 and 24576 DF,  p-value: < 2.2e-16
```

Poisson Regression

```r
model7 <- glm(scored ~ xG_diff + xGA_diff + npxGD,
              family = poisson,
              data = understat_per_game)
summary(model7)
```

```
##
## Call:
## glm(formula = scored ~ xG_diff + xGA_diff + npxGD, family = poisson,
##     data = understat_per_game)
##
## Coefficients:
##              Estimate Std. Error  z value Pr(>|z|)
## (Intercept)  0.042440   0.006556    6.473 9.58e-11 ***
## xG_diff     -0.519748   0.004567 -113.811  < 2e-16 ***
## xGA_diff     0.010185   0.006298    1.617    0.106
```

```
## npxGD        0.323007   0.004415   73.160  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 31556  on 24579  degrees of freedom
## Residual deviance: 12892  on 24576  degrees of freedom
## AIC: 55820
##
## Number of Fisher Scoring iterations: 5
```

Negative Binomial Regression

```r
library(MASS)
```

```
##
## Attaching package: 'MASS'

## The following object is masked from 'package:dplyr':
##
##     select
```

```r
library(MASS)

model8 <- glm.nb(scored ~ xG_diff + xGA_diff + npxGD,
                 data = understat_per_game,
                 control = glm.control(maxit = 100))
```

```
## Warning in theta.ml(Y, mu, sum(w), w, limit = control$maxit, trace =
## control$trace > : iteration limit reached
## Warning in theta.ml(Y, mu, sum(w), w, limit = control$maxit, trace =
## control$trace > : iteration limit reached
```

```r
summary(model8)
```

```
##
## Call:
## glm.nb(formula = scored ~ xG_diff + xGA_diff + npxGD, data = understat_per_game,
##     control = glm.control(maxit = 100), init.theta = 1089805372000,
##     link = log)
##
## Coefficients:
##              Estimate Std. Error  z value Pr(>|z|)
## (Intercept)  0.042440   0.006556    6.473 9.58e-11 ***
## xG_diff     -0.519748   0.004567 -113.811  < 2e-16 ***
## xGA_diff     0.010185   0.006298    1.617    0.106
## npxGD        0.323007   0.004415   73.160  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## (Dispersion parameter for Negative Binomial(1.089805e+12) family taken to be 1)
##
##     Null deviance: 31559  on 24579  degrees of freedom
## Residual deviance: 12892  on 24576  degrees of freedom
## AIC: 55926
##
## Number of Fisher Scoring iterations: 1
##
##
##               Theta:  1.089805e+12
##           Std. Err.:  219619404650
## Warning while fitting theta: iteration limit reached
##
##  2 x log-likelihood:  -55916.16
```

Ridge Regression

```r
library(glmnet)
```

```
## Loading required package: Matrix
```

```
##
## Attaching package: 'Matrix'
```

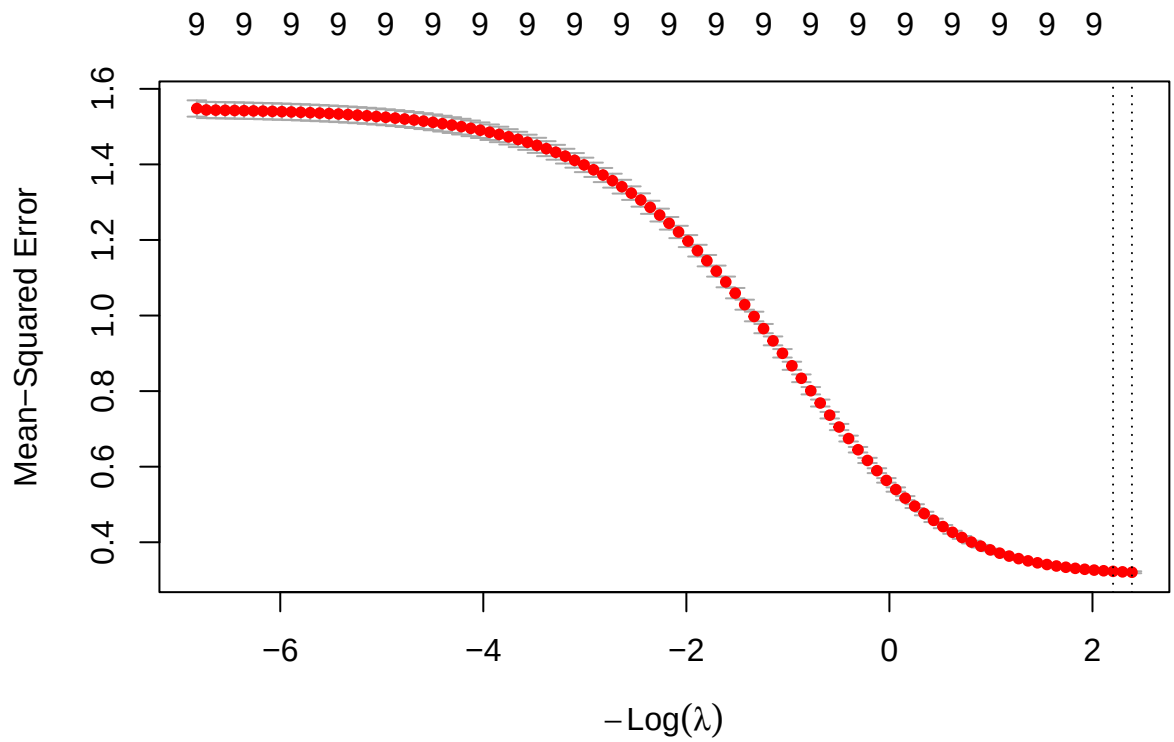
```
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
```

```
## Loaded glmnet 4.1-10
```

```r
x_ridge <- model.matrix(scored ~ xG_diff + xGA_diff + npxGD + ppda_coef +
                          oppda_coef + ppda_att + ppda_def + oppda_att + oppda_def,
                        data = understat_per_game)[, -1]

y_ridge <- understat_per_game$scored

model9 <- cv.glmnet(x_ridge, y_ridge, alpha = 0)
plot(model9)
```

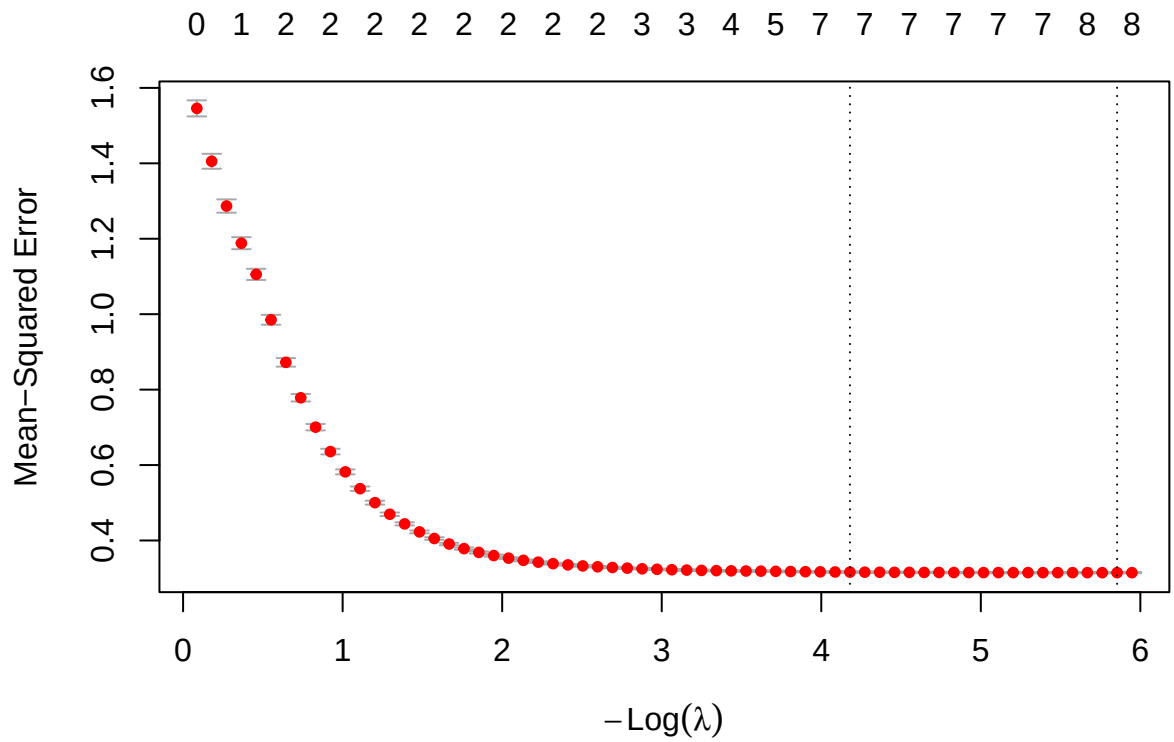



```r
coef(model9, s = "lambda.min")
```

```
## 10 x 1 sparse Matrix of class "dgCMatrix"
##                 lambda.min
## (Intercept)  1.0506770806
## xG_diff     -0.8977684189
## xGA_diff     0.0222323128
## npxGD        0.4708830294
## ppda_coef    0.0037166849
## oppda_coef   0.0059095712
## ppda_att     0.0002209081
## ppda_def     0.0007633308
## oppda_att    0.0004815640
## oppda_def   -0.0011894021
```

LASSO Regression

```r
model10 <- cv.glmnet(x_ridge, y_ridge, alpha = 1)
plot(model10)
```

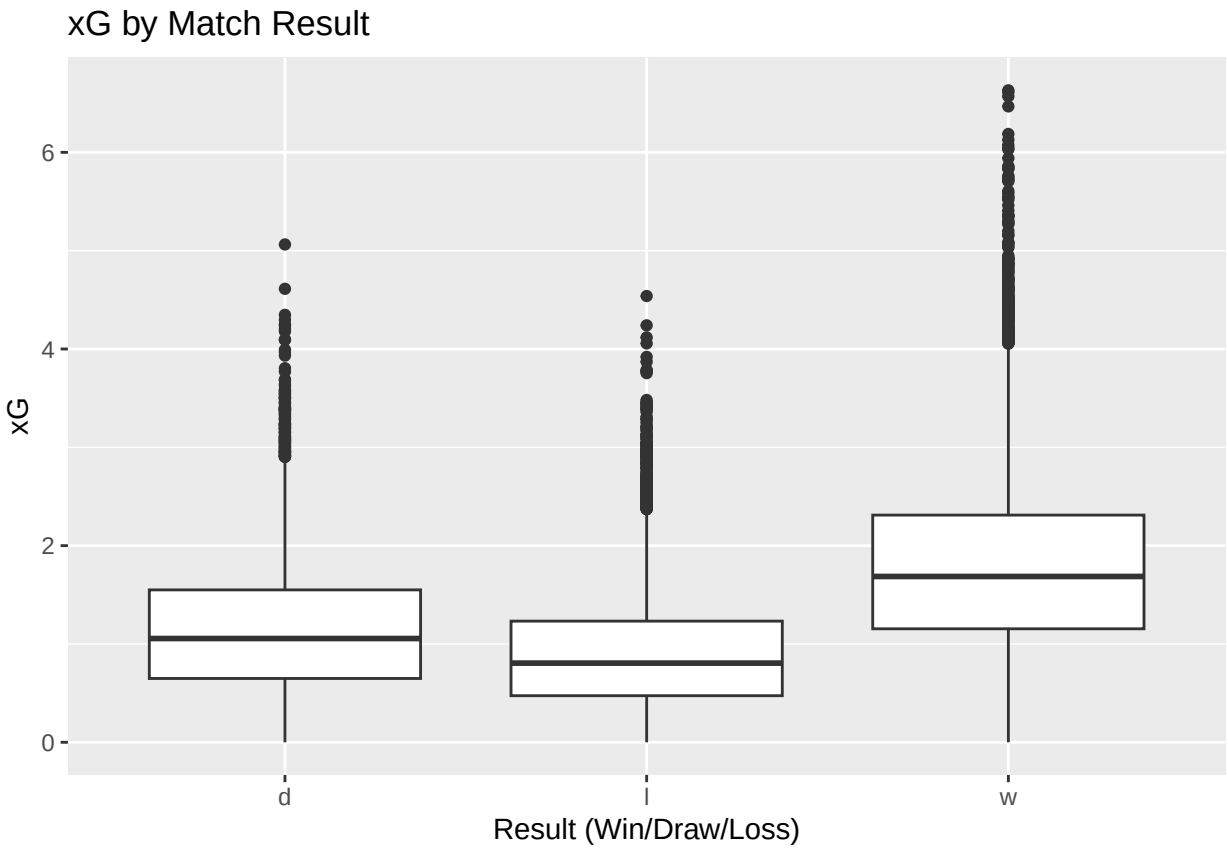



```r
coef(model10, s = "lambda.min")
```

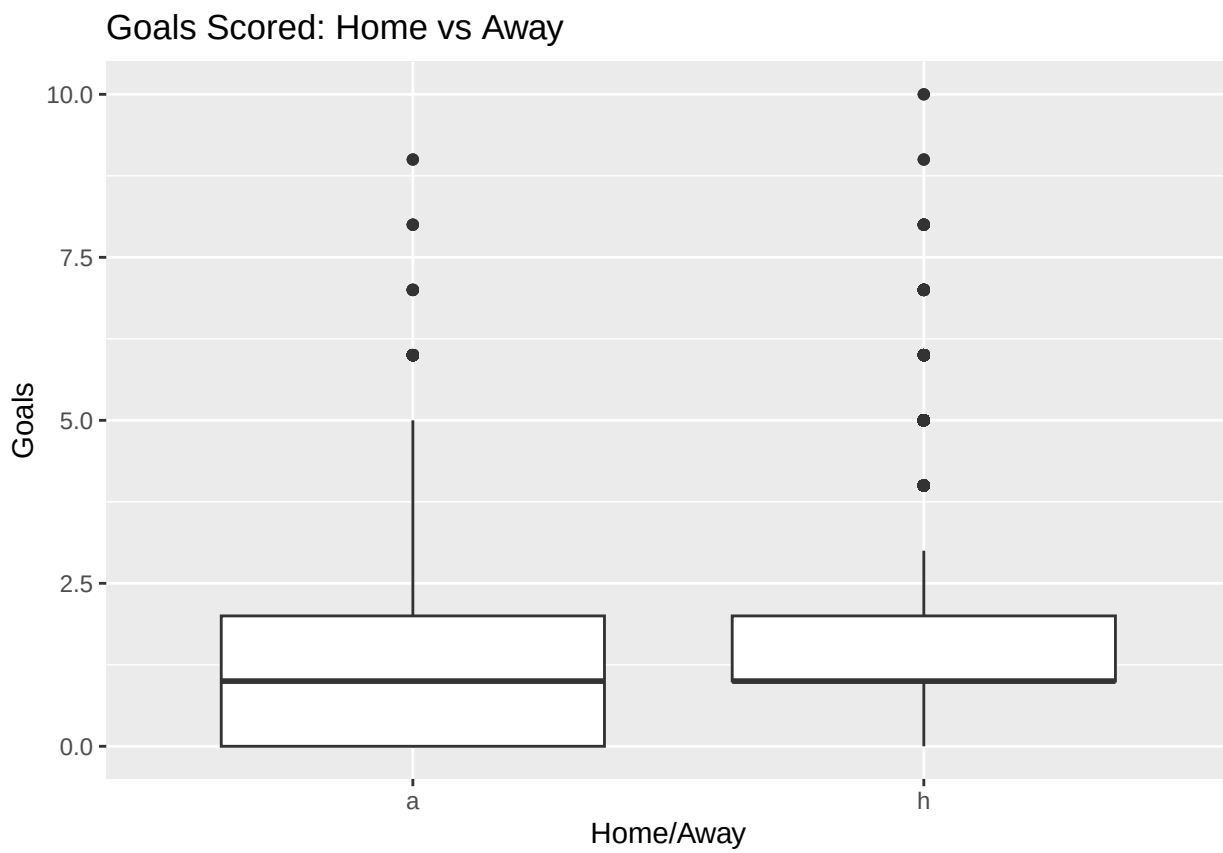
```
## 10 x 1 sparse Matrix of class "dgCMatrix"
##                 lambda.min
## (Intercept)  1.0720944615
## xG_diff     -0.9606302422
## xGA_diff     0.0207384361
## npxGD        0.5088695533
## ppda_coef    0.0039092139
## oppda_coef   0.0062634170
## ppda_att     0.0002467372
## ppda_def     .
## oppda_att    0.0003017660
## oppda_def   -0.0001179634
```

Box Plot

```r
ggplot(understat_per_game, aes(x = result, y = xG)) +
  geom_boxplot() +
  labs(title = "xG by Match Result",
       x = "Result (Win/Draw/Loss)",
       y = "xG")
```



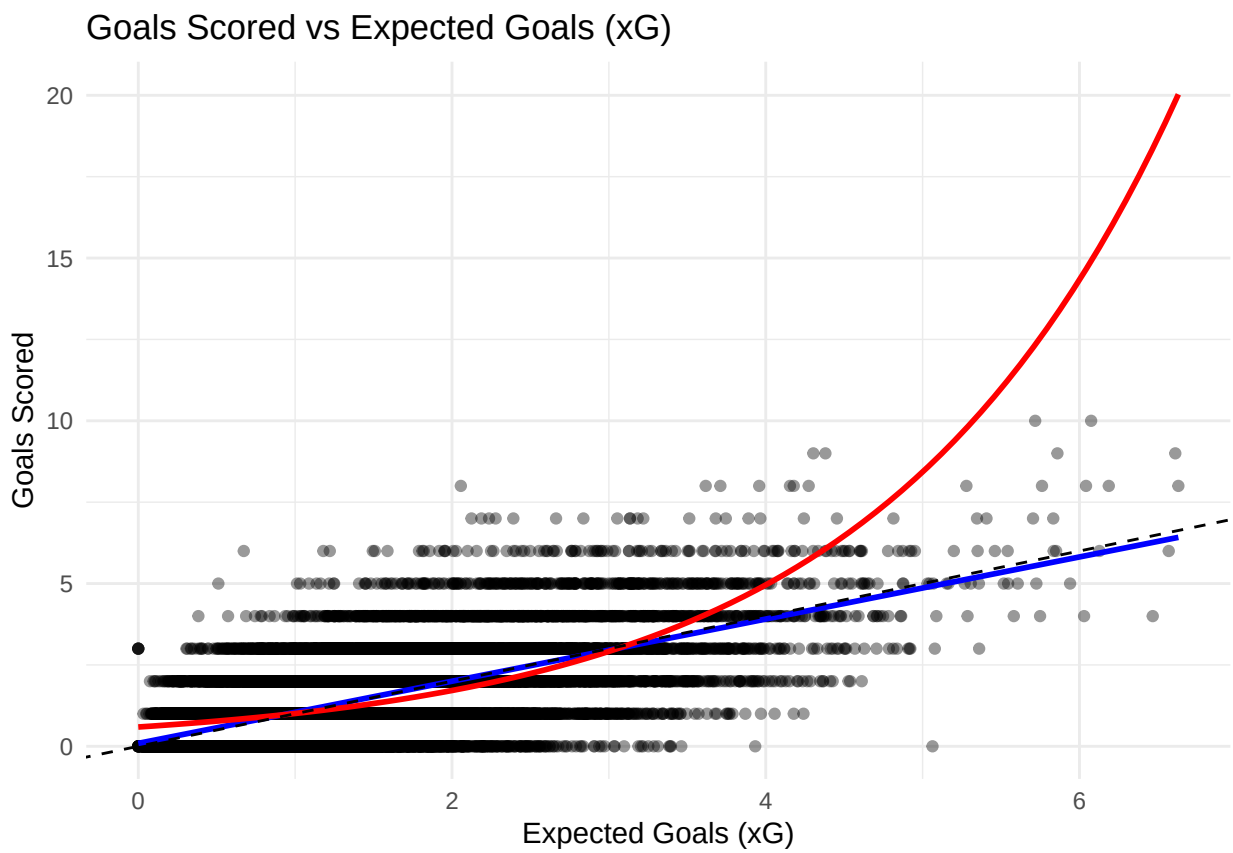

```
ggplot(understat_per_game, aes(x = h_a, y = scored)) +
  geom_boxplot() +
  labs(title = "Goals Scored: Home vs Away",
       x = "Home/Away",
       y = "Goals")
```



ScatterPlot

```
ggplot(understat_per_game, aes(x = xG, y = scored)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = FALSE, color = "blue") +
  geom_smooth(method = "glm", method.args = list(family = "poisson"), se = FALSE, color = "red") +
  labs(title = "Goals Scored vs Expected Goals (xG)",
       x = "Expected Goals (xG)",
       y = "Goals Scored") +
  theme_minimal() +
  geom_abline(intercept = 0, slope = 1, linetype =2, color = "black")
```

```
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
```



This figure shows two different models, predicting goals scored from expected goals. The blue is linear regression and red Poisson regression. When expected goals is less than 3, the two models will give the same prediction. When expected goals is greater than 3, poisson gives you a higher regression, and the dashe line is y=x line (For comparison).